Blob detection with OpenCL

Parallel programming project

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Introduction



Figure 1: Robotex football field

Tresholding



Figure 2: Source image

Tresholding

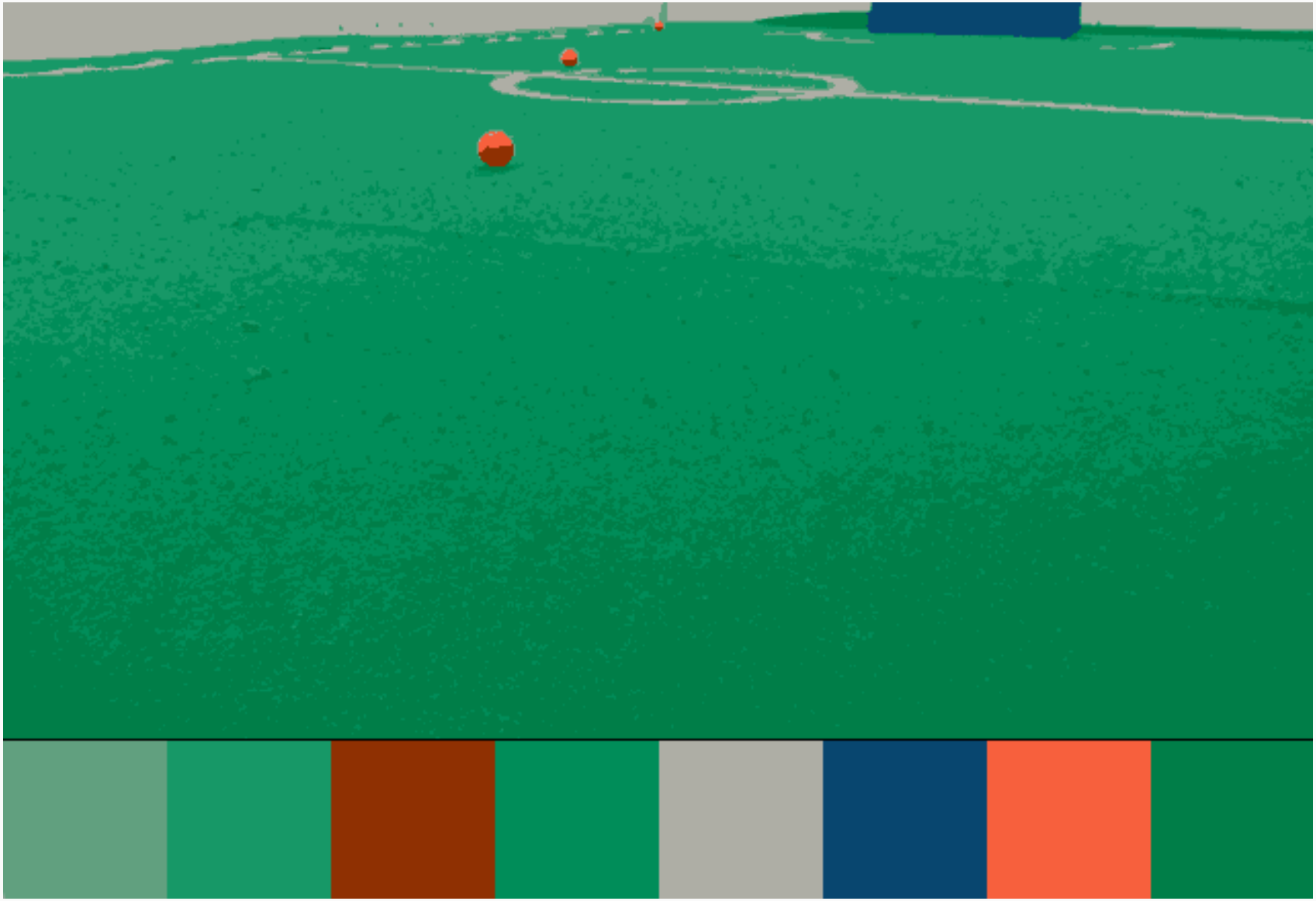


Figure 3: Tresholding result

Tresholding

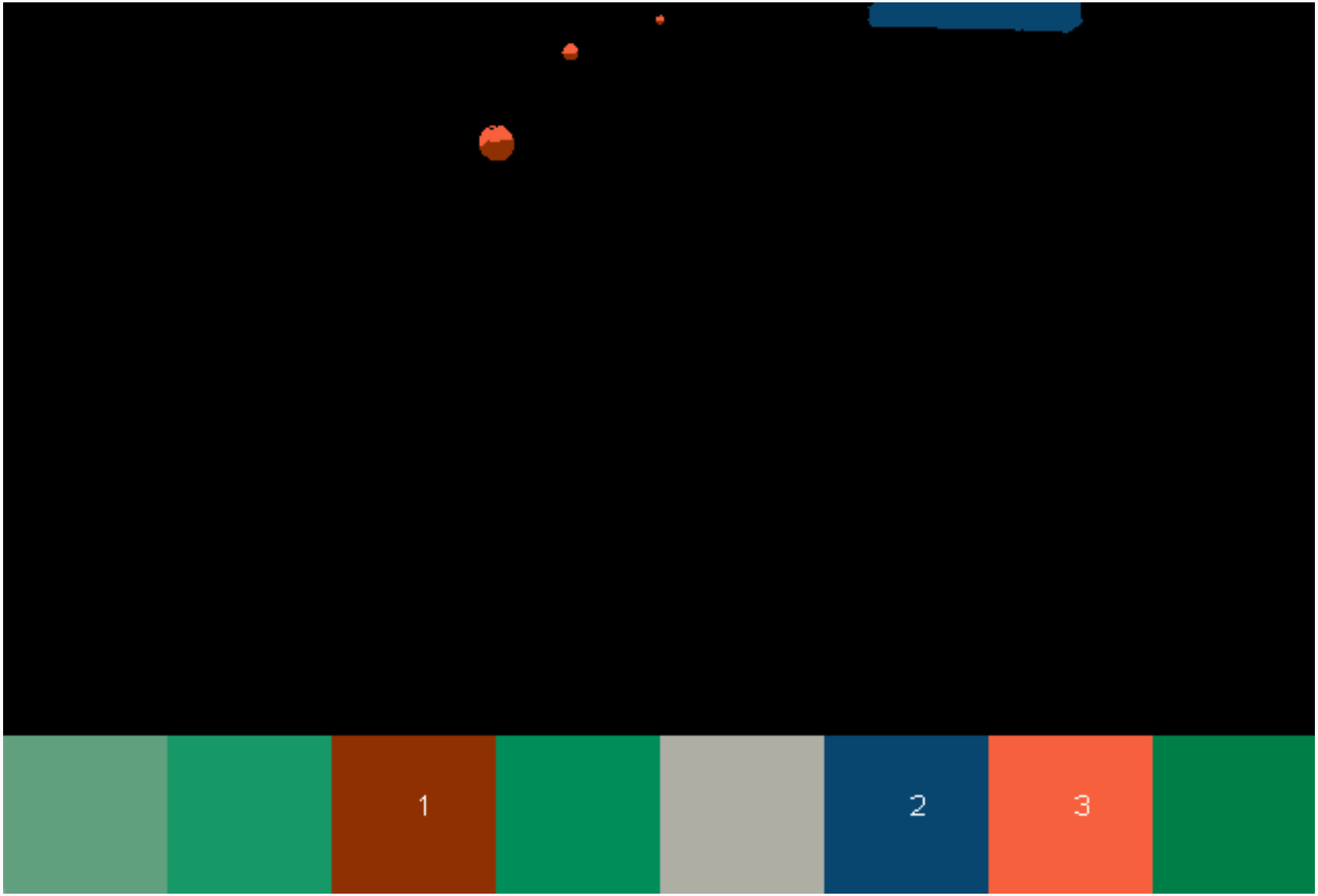


Figure 4: Selecting colors of interest

Blob detection

- Pixels are assigned into clusters
- Sequential same-colored pixels on a row are connected into chains
- Connected chains on adjacent rows are turned into blobs

Blob detection

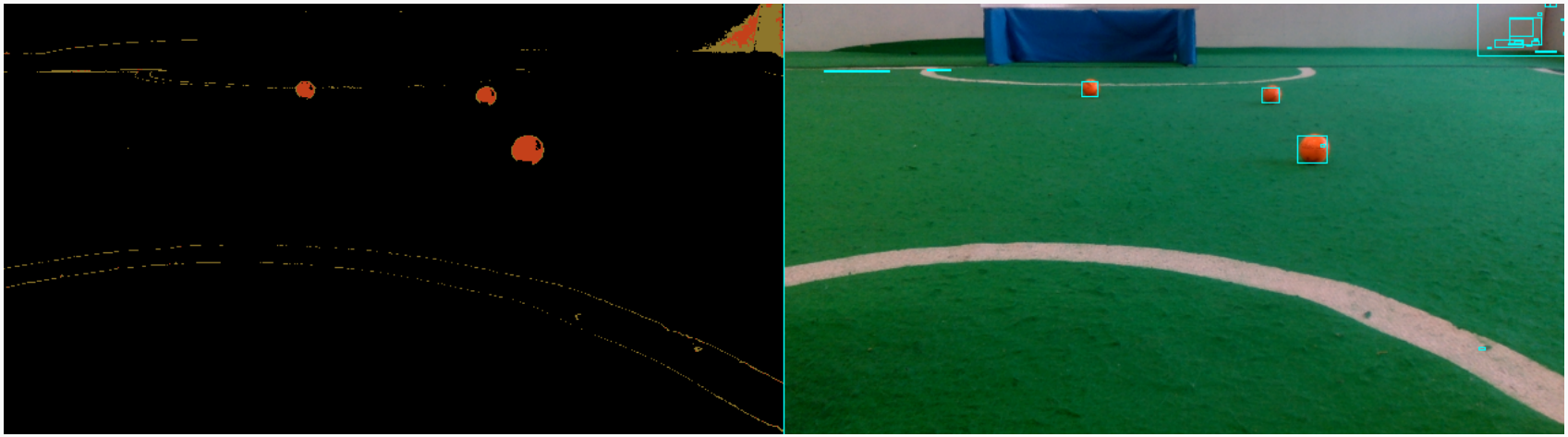


Figure 5: Detecting balls

Discerning similar colors



Figure 6: Objects with similar colors

Discerning similar colors

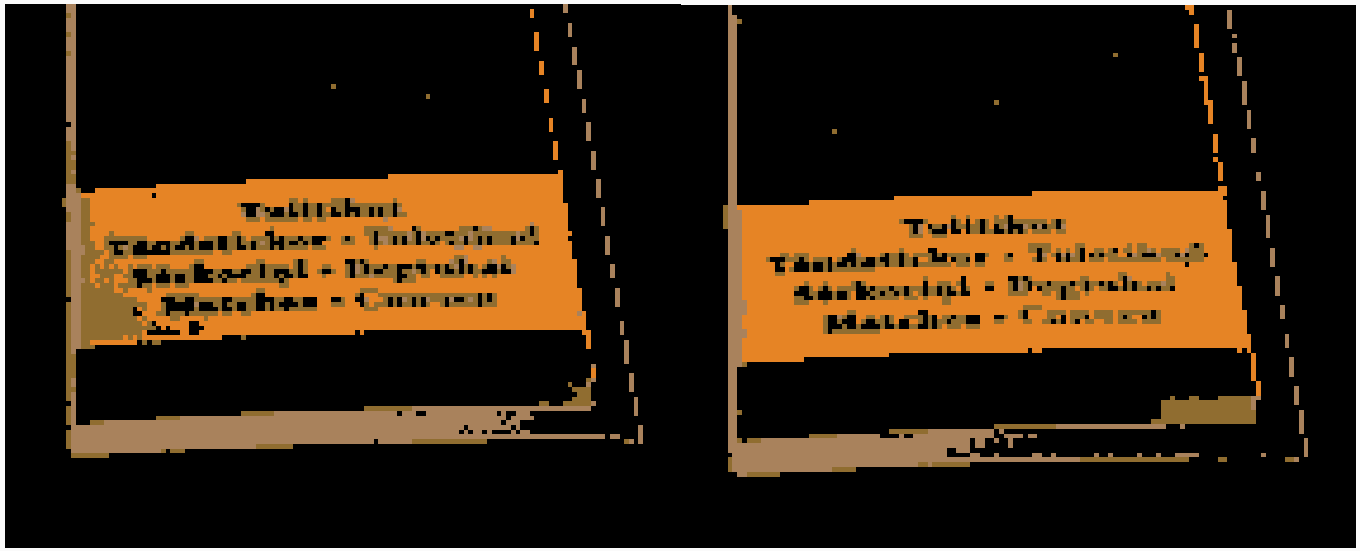


Figure 7: The blob on the right is more pure

Summary

- Learned to use OpenCL to perform parallel image processing
- Good starting point for Robotex 2017

Questions?

Backup slides

Sometimes, it is useful to add slides at the end of your presentation to refer to during audience questions.

The best way to do this is to include the `appendixnumberbeamer` package in your preamble and call `\appendix` before your backup slides.

metropolis will automatically turn off slide numbering and progress bars for slides in the appendix.

References i


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